

Week 10: Time-Varying Treatments & Marginal Structural Models

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Learning Objectives

By the end of this module, you will be able to:

- **Explain** treatment–confounder feedback and why standard adjustment fails (Hernán & Robins, *What If*, §19–20).
- **State and justify** the three core identification assumptions—sequential exchangeability, positivity, and consistency—and diagnose threats to each.
- **Derive and compute** stabilized inverse probability weights (IPWs) for multi-time treatments (Robins, Hernán & Brumback, 2000).
- **Estimate** parameters of a marginal structural model (MSM) with weighted GEE/GLM and **interpret** them as effects of sustained strategies.
- **Compare and contrast** MSM/IPW with the g-formula and structural nested models; articulate when each is advantageous (Neal, *Intro to CI*, Ch. 10–11).
- **Evaluate diagnostics** (weight distribution, truncation, covariate balance) and implement sensitivity checks.

R Packages

```
pacman::p_load(  
  dagitty, # causal DAGs  
  ggdag, # DAG plotting  
  dplyr, # wrangling  
  tidyr, # reshaping  
  tibble, # tibbles  
  purrr, # iteration  
  ggplot2, # visualization  
  geepack, # GEE for MSMs  
  survey, # weighted GLMs  
  WeightIt, # MSM weights  
  cobalt, # balance diagnostics  
  sandwich, # robust vcov  
  lmtest # coeftest  
)
```

1 Comprehensive Topic Explanation

1.1 The Longitudinal Setup

We observe for each individual i across $k = 0, \dots, K$:

- Time-varying covariates $\bar{L}_k = (L_0, \dots, L_k)$,
- Treatments $\bar{A}_k = (A_0, \dots, A_k)$,
- Outcome Y at $K+1$.

Treatment-confounder feedback: L_k both (i) confounds $A_k \rightarrow Y$ and (ii) lies on the pathway $A_{k-1} \rightarrow L_k \rightarrow Y$. Adjusting for L_k in a standard outcome regression blocks a piece of the very effect we want (mediated path), inducing bias or sign reversal (Hernán & Robins §19).

1.2 Target Estimands & MSMs

We shift from “effect of A_k given L_k ” to the **marginal effect of a sustained strategy**, e.g., “treat whenever biomarker $> x$.” MSMs parameterize the **marginal** counterfactual mean under a regime:

$$\mathbb{E}\{Y^{\bar{a}}\} = m(\bar{a}; \beta),$$

e.g., a logit-linear MSM for two time points:

$$\text{logit}\{\mathbb{E}(Y^{a_0 a_1})\} = \beta_0 + \beta_1 a_0 + \beta_2 a_1 + \beta_3 a_0 a_1.$$

Parameters β are interpreted as **causal contrasts of sustained histories**.

1.3 Identification Assumptions (Hernán & Robins §19; Neal Ch. 10)

- **Sequential exchangeability:** $Y^{\bar{a}} \perp\!\!\!\perp A_k \mid \bar{L}_k, \bar{A}_{k-1}$ for all k .
Given observed history, treatment is as-if random.
- **Positivity:** $\Pr(A_k = a_k \mid \bar{L}_k = \bar{l}_k, \bar{A}_{k-1} = \bar{a}_{k-1}) > 0$ whenever $\Pr(\bar{L}_k = \bar{l}_k, \bar{A}_{k-1} = \bar{a}_{k-1}) > 0$.
- **Consistency:** If an individual follows $\bar{A} = \bar{a}$, then $Y = Y^{\bar{a}}$.

1.4 IPW Construction & Stabilized Weights

To break confounding by $\bar{L}_k$, define **stabilized weight** for individual i :

$$SW_i = \prod_{k=0}^K \frac{\Pr(A_{ik} = a_{ik} \mid \bar{A}_{i,k-1})}{\Pr(A_{ik} = a_{ik} \mid \bar{A}_{i,k-1}, \bar{L}_{ik})}.$$

Intuition: **upweight** histories that were unlikely to get the treatment they received *given* their covariates; **downweight** those that were likely. In the weighted pseudo-population, $\bar{L}_k \not\rightarrow A_k$ by design; MSM

parameters can be fit by weighted regression/GEEs (Robins, Hernán & Brumback, 2000).

1.5 Estimation Strategies

- **IPTW of the marginal mean:** estimate $\mathbb{E}(Y^{\bar{a}})$ by weighting each unit by the inverse probability of their observed treatment path (restricted to those following $\bar{a}$ when estimating that regime).
- **Weighted GLM/GEE (common in practice):** regress Y on functions of $\bar{A}$ per the MSM using SW as weights; use robust (sandwich) SEs.
- **Variance:** robust sandwich; cluster on ID for repeated measures.
- **Censoring:** incorporate **censoring weights** by multiplying treatment weights with inverse probabilities of remaining uncensored (Hernán & Robins §12, §19).

1.6 MSMs Vs G-formula Vs SNMs (compare–contrast)

- **MSM + IPW:** Targets marginal regime effects; robust to outcome model misspecification but sensitive to treatment model/positivity & weight instability.
- **G-formula (parametric g-comp):** Models $Y \mid \text{history}$ and simulates under regimes. Sensitive to outcome model misspecification; no unstable weights but more modeling.
- **SNMs/g-estimation:** Alternative parameterization focusing on treatment blip effects; useful with time-varying treatments and confounding; often less familiar operationally.

Decision Guide

When choose what? - Strong domain knowledge to specify $Y \mid \text{history}$ → **g-formula**. - Cleaner treatment model & desire marginal interpretation → **MSM**. - Focus on dynamic blip functions & structural nesting → **SNM**.

Citations: Hernán & Robins §12, §19–20; Neal Ch. 10–11; Robins (1986, 1997, 2000); VanderWeele Ch. 19.

2 Intuitive Clarifications of Challenging Ideas

💡 Learning Tip

Think of **IPW** as “re-queuing the study” so that—conditional on past treatments—the *next* treatment looks like a coin flip, independent of the covariates that used to influence it.

Visual metaphor: - Pipes and valves. Past treatment opens/closes valves that change L_k (flow rate). Those L_k then steer future treatment. IPW temporarily “disconnects” the $L_k \rightarrow A_k$ pipe so we can see the net effect of opening the treatment valves over time.

Common misconceptions (and fixes): - “*I adjusted for L_k in an outcome regression, so I controlled confounding.*” **No**—you also block mediated effects of earlier A through L_k , biasing the target marginal effect (Hernán & Robins §19). - “*Large weights just mean more precision.*” **No**—they usually signal near-positivity violations; consider truncation, coarsening, or redefining the target regime/support. - “*Stabilized weights always fix everything.*” They reduce variance, not bias from misspecified models or violations of assumptions.

DAG for treatment–confounder feedback (drawn below) clarifies the double role of L_1 .

```
dag_basic <- dagitty(  
  "  
  dag {  
    L0; A0; L1; A1; Y  
    L0 → A0  
    L0 → L1  
    A0 → L1  
    L1 → A1  
    A0 → Y  
    A1 → Y  
  }  
  )
```

```

  L1 → Y
}"
)
ggdag(dag_basic, layout = "nicely") + theme_dag()

```

3 Simple, Hand-Calculation Numerical Example(s)

3.1 Four-person Toy Example (IPW mechanics)

Two times ($k = 0, 1$), all binary. Data:

ID	L0	A0	L1	A1	Y
1	0	0	0	0	0
2	0	1	1	1	1
3	1	0	1	0	0
4	1	1	0	1	1

Step A (fit by hand): empirical treatment probabilities

- $P(A_0=1 | L_0=0) = 1/2, P(A_0=1 | L_0=1) = 1/2$
- $P(A_1=1 | A_0=1, L_1=0) = 1, P(A_1=1 | A_0=0, L_1=0) = 0,$
 $P(A_1=1 | A_0=1, L_1=1) = 1, P(A_1=1 | A_0=0, L_1=1) = 0$ (from table)

Numerators (stabilized) use reduced models: - $P(A_0=1) = 1/2$; $P(A_1=1 | A_0)$ equals 1 when $A_0=1$, and 0 when $A_0=0$ (in this tiny table).

Step B (compute each SW_i): Example, ID 2 has $(A_0, A_1) = (1, 1)$ with $(L_0, L_1) = (0, 1)$:

$$SW_2 = \frac{P(A_0=1)}{P(A_0=1 | L_0=0)} \cdot \frac{P(A_1=1 | A_0=1)}{P(A_1=1 | A_0=1, L_1=1)} = \frac{1/2}{1/2} \cdot \frac{1}{1} = 1.$$

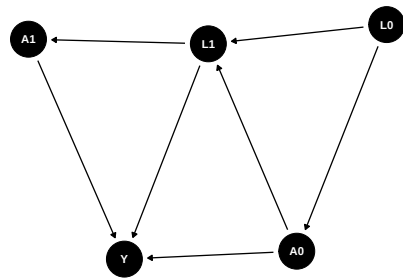


Figure 1: Minimal DAG with treatment-confounder feedback at two times

Do likewise for IDs 1,3,4 (here, all weights turn out 1 due to deterministic A_1 within strata). This *illustrates* stabilized weight definition even if trivial.

Step C (weighted mean under regime “always treat”) To estimate $\mathbb{E}(Y^{1,1})$, keep only those with $(A_0, A_1) = (1, 1)$ (IDs 2 and 4), and compute the **weighted** mean of Y : $(1 \cdot 1 + 1 \cdot 1) / (1 + 1) = 1$. For “never treat,” use $(0, 0)$ (ID 1 only): mean = 0 (ID 3 has $(0, 0)$? No; it’s $(0, 0)$ at A , yes— $Y = 0$). Estimated effect (risk difference) $\approx 1 - 0 = 1$ in this toy.

! □ Key Assumption

This hand example has **deterministic** A_1 given (A_0, L_1) , making positivity fragile. Real data rarely behave so cleanly—expect non-trivial weights and perform diagnostics.

3.2 Scaffolding Extension (non-trivial numerics)

Modify the table so A_1 sometimes flips in $(A_0=1, L_1=0)$ with probability 0.7 and in $(A_0=1, L_1=1)$ with 0.6. Recompute denominators, numerators (e.g., $P(A_1=1 \mid A_0=1) = 0.65$), then each SW_i . Observe how **rare** patterns get **larger** weights.

4 R Code Examples & Interpretation

4.1 DAGs for Feedback and for Censoring (advanced)

```
dag_censor <- dagitty(
  "
dag {
  L0; A0; L1; A1; C1; Y
  L0 → A0
  L0 → L1
  A0 → L1
  L1 → A1
```

```

L1 → C1
A0 → Y
A1 → Y
L1 → Y
C1 → Y
}"
)
ggdag(dag_censor, layout = "nicely") + theme_dag()

```

Interpretation: C_1 (censoring/loss to follow-up) depends on L_1 ; we need **censoring weights** in addition to treatment weights when Y is missing for some.

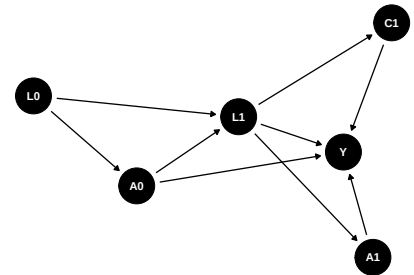


Figure 2: MSM DAG with time-varying confounders and censoring

4.2 Basic Simulation: Two time Points, Binary Outcome

```

set.seed(123)
n <- 4000
L0 <- rbinom(n, 1, 0.5)
A0 <- rbinom(n, 1, plogis(-0.5 + 1.0 * L0))
L1 <- rbinom(n, 1, plogis(-0.2 + 1.2 * A0 + 0.6 * L0))
A1 <- rbinom(n, 1, plogis(-0.8 + 1.6 * L1 + 0.3 * A0))
linY <- -0.3 + 0.9 * A0 + 1.1 * A1 + 0.5 * L1
Y <- rbinom(n, 1, plogis(linY))
dat <- tibble(id = 1:n, L0, A0, L1, A1, Y)

# Denominator Models (rich)
pA0_den <- glm(A0 ~ L0, family = binomial, data =
dat)$fitted.values
pA1_den <- glm(A1 ~ A0 + L0 + L1, family = binomial,
data = dat)$fitted.values

# Numerator Models (reduced)
pA0_num <- glm(A0 ~ 1, family = binomial, data =
dat)$fitted.values
pA1_num <- glm(A1 ~ A0, family = binomial, data =
dat)$fitted.values

```


Table 2: Simulated weights with truncation and normalization

```
alpha <- 0.01
lower <- quantile(dat$sw, alpha)
upper <- quantile(dat$sw, 1 - alpha)
dat <- dat %>%
  mutate(
    sw_trunc = pmin(pmax(sw, lower), upper),
    sw_norm = sw / mean(sw),
    swt_norm = sw_trunc / mean(sw_trunc)
  )
summary(dat$sw)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5774	0.7146	0.9524	1.2407	1.5162	2.4374

```
summary(dat$sw_trunc)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5774	0.7146	0.9524	1.2407	1.5162	2.4374

```
dat <- dat %>% mutate(sw = (pA0_num / pA0_den) * (pA1_num / pA1_den))

ggplot(dat, aes(sw)) +
  geom_histogram(bins = 50)
```

Interpretation. A tight weight distribution suggests OK overlap; heavy tails signal potential positivity issues.

Optional truncation & normalization:

4.3 MSM Estimation (weighted GLM/GEE) and Naïve Comparison

Compare–contrast: The naïve model adjusts for L_1 (a mediator of A_0), biasing $\hat{\beta}_{A_0}$. The MSM targets **marginal** effects; coefficients approximate the causal log-odds effects of treatment at each time.

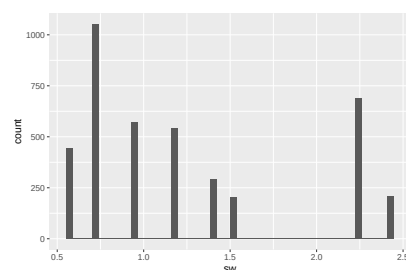


Figure 3: Distribution of stabilized weights (basic simulation)

Table 3: Naive and MSM Weighted GEE Estimates with Truncation

```
# Naïve (condition on L1 incorrectly)
naive <- glm(Y ~ A0 + A1 + L1, family = binomial, data =
dat)
summary(naive)
```

Call:

```
glm(formula = Y ~ A0 + A1 + L1, family = binomial, data = dat)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.28018	0.06440	-4.351	1.36e-05 ***
A0	0.96195	0.08210	11.717	< 2e-16 ***
A1	1.20951	0.08284	14.600	< 2e-16 ***
L1	0.41554	0.08442	4.922	8.56e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 4676.0 on 3999 degrees of freedom
Residual deviance: 4065.6 on 3996 degrees of freedom
AIC: 4073.6

Number of Fisher Scoring iterations: 4

```
# MSM with Weighted GEE (independence Working
correlation)
msm <- geeglm(
  Y ~ A0 + A1,
  id = id,
  weights = sw_trunc,
  data = dat,
  family = binomial,
  corstr = "independence"
)
summary(msm)
```

Call:

```
geeglm(formula = Y ~ A0 + A1, family = binomial, data = dat,
weights = sw_trunc, id = id, corstr = "independence")
```

Coefficients:

	Estimate	Std.err	Wald	Pr(> W)
(Intercept)	-0.20026	0.06309	10.08	0.0015 **
A0	1.04836	0.08892	138.99	<2e-16 ***
A1	1.31881	0.08662	231.80	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = independence

Estimated Scale Parameters:

4.4 Diagnostics: Balance with WeightIt + cobalt (scaffolded)

```
# WeightIt for MSM (two time points)
w_msm <- weightitMSM(
  list(A0 ~ L0, A1 ~ A0 + L0 + L1),
  data = dat,
  method = "ps",
  stabilize = TRUE
)
all.equal(as.numeric(w_msm$weights), dat$sw, tolerance =
1e-6) # similar
```

```
[1] "names for current but not for target"
[2] "Mean relative difference: 0.6144"
```

```
# Balance Diagnostics at Each K Using Cobalt
bal <- bal.tab(w_msm, m.threshold = .1, un = TRUE)
bal
```

Balance summary across all time points

	Times	Type	Max.Diff.Un	Max.Diff.Adj	M.Threshold
L0	1, 2	Binary	0.258	0.004	Balanced, <0.1
A0	2	Binary	0.184	0.185	Not Balanced, >0.1
L1	2	Binary	0.386	0.049	Balanced, <0.1

Balance tally for mean differences

	count
Balanced, <0.1	2
Not Balanced, >0.1	1

Variable with the greatest mean difference

Variable	Max.Diff.Adj	M.Threshold
A0	0.185	Not Balanced, >0.1

Effective sample sizes

```

- Time 1
      Control Treated
Unadjusted   1994   2006
Adjusted     1569   1674
- Time 2
      Control Treated
Unadjusted   1676   2324
Adjusted     1330   1910

```

```
love.plot(bal)
```

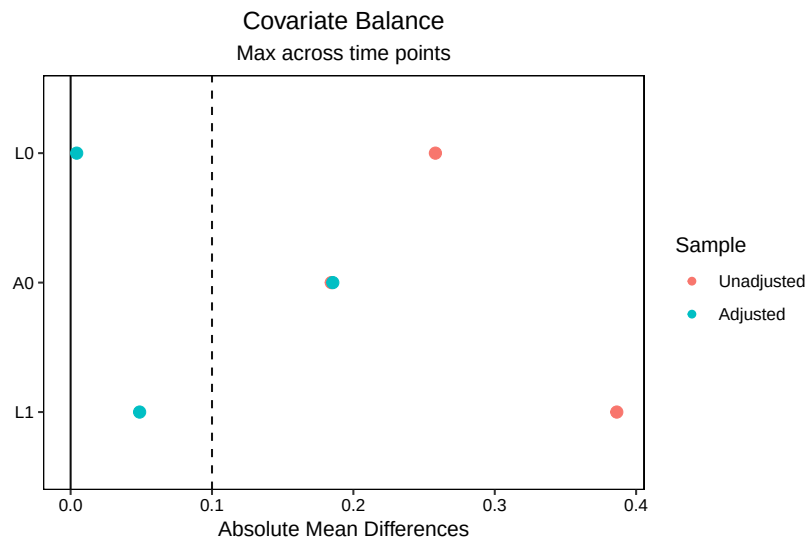


Figure 4: WeightIt Stabilized Weights for MSM

Interpretation. Covariate balance (standardized mean differences) should improve across treatment groups after weighting, **at the times they act as confounders**.

4.5 Advanced Simulation: Add Censoring & Censored Weights

Note. For time-to-event outcomes, use `survival::coxph` with weights (and robust variance).

Table 4: Simulated weights with censoring

```
set.seed(1)
dat2 <- dat >
  mutate(C1 = rbinom(n(), 1, plogis(-2 + 1.3 * L1 - 0.6
    * A1))) # 1 = censored
# Denominator and Numerator Models for Censoring (prob
of *not* censored)
pC1_den <- glm(
  I(1 - C1) ~ A0 + A1 + L0 + L1,
  family = binomial,
  data = dat2
)$fitted.values
pC1_num <- glm(
  I(1 - C1) ~ A0 + A1,
  family = binomial,
  data = dat2
)$fitted.values
dat2 <- dat2 >
  mutate(
    sc = (pC1_num / pC1_den),
    swc = sw * sc,
    swc_trunc = pmin(pmax(swc, lower), upper)
  )

# Fit MSM among Uncensored with Combined Weights
msm_c <- geeglm(
  Y ~ A0 + A1,
  id = id,
  weights = swc_trunc,
  data = filter(dat2, C1 == 0),
  family = binomial,
  corstr = "independence"
)
summary(msm_c)
```

Call:

```
geeglm(formula = Y ~ A0 + A1, family = binomial, data = filter(dat2,
  C1 == 0), weights = swc_trunc, id = id, corstr = "independence")
```

Coefficients:

	Estimate	Std.err	Wald	Pr(> W)
(Intercept)	-0.1903	0.0686	7.7	0.0055 **
A0	1.0714	0.0964	123.5	<2e-16 ***
A1	1.2757	0.0929	188.4	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

13

Correlation structure = independence

Estimated Scale Parameters:

	Estimate	Std.err
(Intercept)	1	0.0552

Number of clusters: 3182 Maximum cluster size: 1

5 Hands-On R Lab

Purpose: Practice computing stabilized weights, diagnosing overlap, and estimating MSMs. **Pre-lab reading (flipped):** Hernán & Robins §19–20; Neal Ch. 10. **Outcomes:** You will implement IPW for a 2-time MSM, compare with a naïve model, and reflect on assumptions and diagnostics.

Lab Objectives

1. Simulate a dataset with treatment–confounder feedback.
2. Draw the causal DAG and mark confounder/mediator roles.
3. Compute stabilized weights (with and without truncation).
4. Fit naïve vs MSM models; interpret differences.
5. Assess weight distributions and covariate balance.
6. (Stretch) Add censoring weights and refit.

Reading Warm-up (2–3 bullets)

- Why does adjusting for L_1 in $Y \sim A_0, A_1, L_1$ bias the effect of A_0 ? (*What If* §19)
- What does the **numerator** in stabilized weights accomplish?

Task 1 — Simulate Data

```
# Skeleton (fill in the ... lines)
set.seed(10)
n <- 3000
L0 <- rbinom(n, 1, 0.5)
A0 <- rbinom(n, 1, plogis(-0.4 + 0.9 * L0))
L1 <- rbinom(n, 1, plogis(-0.1 + 1.1 * A0 + 0.5 * L0))
A1 <- rbinom(n, 1, plogis(-0.6 + 1.4 * L1 + 0.3 * A0))
Y <- rbinom(n, 1, plogis(-0.2 + 0.8 * A0 + 1.0 * A1 +
0.4 * L1))
lab <- tibble(id = 1:n, L0, A0, L1, A1, Y)
```

```
head(lab)
```

Table 5: Simulated dataset for lab

id	L0	A0	L1	A1	Y
1	1	1	1	1	1
2	0	0	0	0	0
3	0	0	1	0	1
4	1	0	1	1	1
5	0	0	1	1	0
6	0	0	1	0	1

Solution:

```
lab <- lab
```

Task 2 — Draw the DAG

```
dag_lab <- dagitty(  
  "  
  dag {  
    L0; A0; L1; A1; Y  
    L0 → A0  
    L0 → L1  
    A0 → L1  
    L1 → A1  
    L1 → Y  
    A0 → Y  
    A1 → Y  
  }"  
)  
ggdag(dag_lab, layout = "nicely") + theme_dag()
```

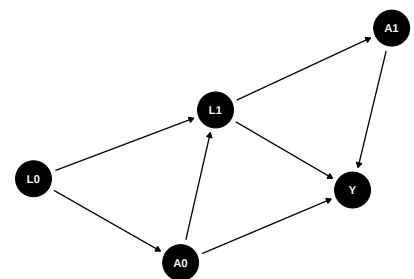


Figure 5: Lab DAG: feedback at two times

Table 6: Lab stabilized weights

```
# Denominators
pA0_d <- glm(A0 ~ L0, family = binomial, data =
lab)$fitted.values
pA1_d <- glm(A1 ~ A0 + L0 + L1, family = binomial, data
= lab)$fitted.values
# Numerators
pA0_n <- glm(A0 ~ 1, family = binomial, data =
lab)$fitted.values
pA1_n <- glm(A1 ~ A0, family = binomial, data =
lab)$fitted.values
lab <- lab %>%
  mutate(sw = (pA0_n / pA0_d) * (pA1_n / pA1_d))
summary(lab$sw)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.627	0.727	0.976	1.173	1.405	2.133

Task 3 — Compute Stabilized Weights

Truncation ($\alpha = 1\%$):

```
alpha <- 0.01
lwr <- quantile(lab$sw, alpha)
upr <- quantile(lab$sw, 1 - alpha)
lab <- lab %>%
  mutate(sw_t = pmin(pmax(sw, lwr), upr))
head(lab)
```

Table 7: Lab stabilized weights with truncation

id	L0	A0	L1	A1	Y	sw	sw_t
1	1	1	1	1	1	0.727	0.727
2	0	0	0	0	0	1.974	1.974
3	0	0	1	0	1	0.976	0.976
4	1	0	1	1	1	0.627	0.627
5	0	0	1	1	0	0.976	0.976

Table 7: Lab stabilized weights with truncation

id	L0	A0	L1	A1	Y	sw	sw_t
6	0	0	1	0	1	0.976	0.976

Task 4 — Fit Naïve Vs MSM

```
naive_lab <- glm(Y ~ A0 + A1 + L1, family = binomial,
data = lab)
msm_lab <- geeglm(
  Y ~ A0 + A1,
  id = id,
  weights = sw_t,
  data = lab,
  family = binomial,
  corstr = "independence"
)
summary(naive_lab)
```

Call:

```
glm(formula = Y ~ A0 + A1 + L1, family = binomial, data = lab)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.3001	0.0768	-3.90	9.4e-05 ***
A0	0.9217	0.0899	10.25	< 2e-16 ***
A1	0.9673	0.0912	10.61	< 2e-16 ***
L1	0.4726	0.0935	5.06	4.3e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 3574.6 on 2999 degrees of freedom

Residual deviance: 3204.3 on 2996 degrees of freedom
AIC: 3212

Number of Fisher Scoring iterations: 4

```
summary(msm_lab)
```

Call:

```
geeglm(formula = Y ~ A0 + A1, family = binomial, data = lab,  
        weights = sw_t, id = id, corstr = "independence")
```

Coefficients:

	Estimate	Std.err	Wald	Pr(> W)	
(Intercept)	-0.2422	0.0742	10.7	0.0011	**
A0	1.0547	0.0955	121.9	<2e-16	***
A1	1.1317	0.0932	147.5	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = independence

Estimated Scale Parameters:

	Estimate	Std.err
(Intercept)	1.01	0.0507

Number of clusters: 3000 Maximum cluster size: 1

Interpretation prompts (write inline answers): - Compare $\hat{\beta}_{A0}$ across models. Which aligns with the data-generating truth? - Why does including L_1 in the naïve model distort the A_0 coefficient?

Task 5 — Diagnose Weights & Balance

```
p1 <- ggplot(lab, aes(sw)) +  
  geom_histogram(bins = 50)  
p2 <- ggplot(lab, aes(sw_t)) +  
  geom_histogram(bins = 50)  
p1  
p2
```

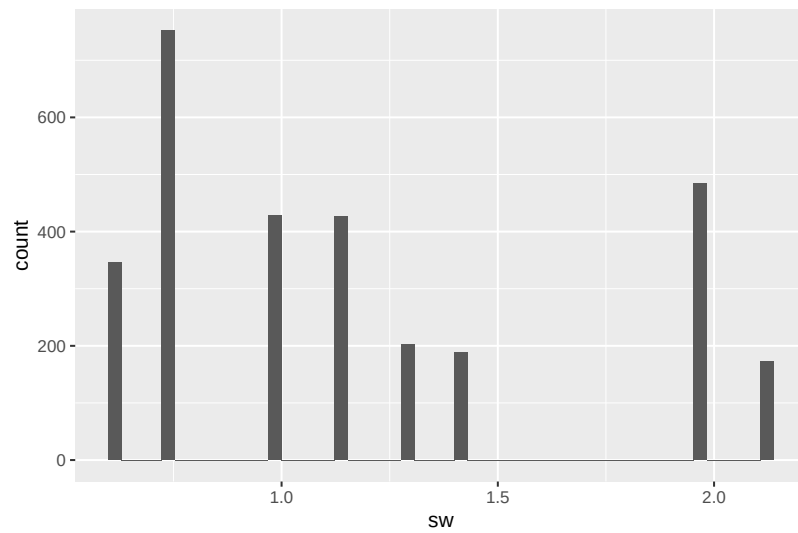


Figure 6: Lab: weight histograms

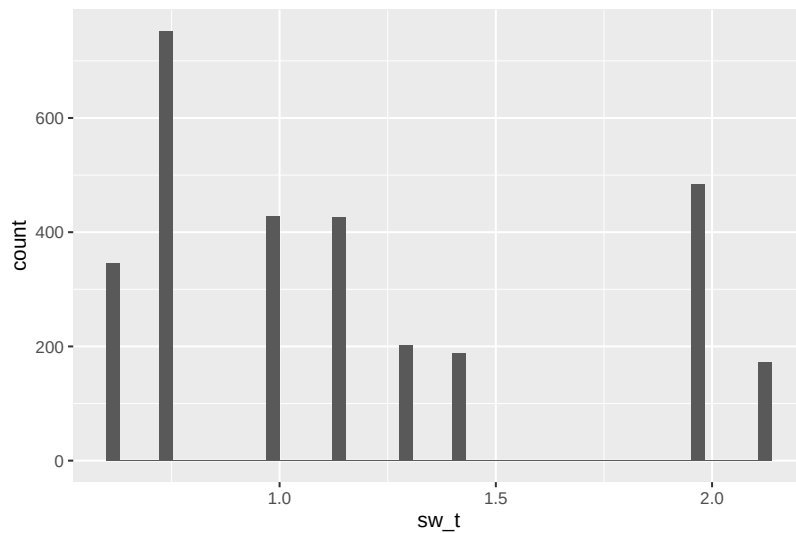


Figure 7: Lab: weight histograms

```
w_msm_lab <- weightitMSM(
  list(A0 ~ L0, A1 ~ A0 + L0 + L1),
  data = lab,
  method = "ps",
  corstr = "independence",
  stabilize = TRUE
)
bal_lab <- bal.tab(w_msm_lab, un = TRUE, m.threshold =
.1)
bal_lab
```

Balance summary across all time points

	Times	Type	Max.Diff.Un	Max.Diff.Adj	M.Threshold
L0	1, 2	Binary	0.236	0.002	Balanced, <0.1
A0	2	Binary	0.129	0.128	Not Balanced, >0.1
L1	2	Binary	0.349	0.028	Balanced, <0.1

Balance tally for mean differences

	count
Balanced, <0.1	2
Not Balanced, >0.1	1

Variable with the greatest mean difference

Variable	Max.Diff.Adj	M.Threshold
A0	0.128	Not Balanced, >0.1

Effective sample sizes

- Time 1

	Control	Treated
Unadjusted	1460	1540
Adjusted	1201	1319

- Time 2

	Control	Treated
Unadjusted	1192	1808
Adjusted	964	1556

```
love.plot(bal_lab)
```

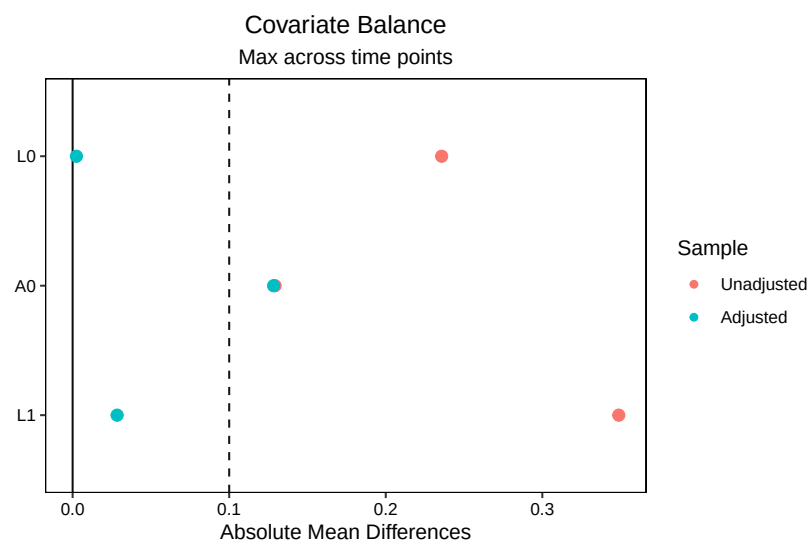


Figure 8: Lab: balance

Task 6 — (Stretch) Add Censoring Weights

```
set.seed(9)
lab2 <- lab %>%
  mutate(C1 = rbinom(n(), 1, plogis(-2 + 1.0 * L1 - 0.5
    * A1))) # 1=censored
pC1_d <- glm(
  I(1 - C1) ~ A0 + A1 + L0 + L1,
  family = binomial,
  data = lab2
)$fitted.values
pC1_n <- glm(I(1 - C1) ~ A0 + A1, family = binomial,
  data = lab2)$fitted.values
lab2 <- lab2 %>%
  mutate(sc = pC1_n / pC1_d, swc = sw_t * sc)
msm_c_lab <- geeglm(
  Y ~ A0 + A1,
  id = id,
  weights = swc,
  data = filter(lab2, C1 == 0),
  family = binomial,
  corstr = "independence"
)
summary(msm_c_lab)
```

Call:

```
geeglm(formula = Y ~ A0 + A1, family = binomial, data = filter(lab2,
  C1 == 0), weights = swc, id = id, corstr = "independence")
```

Coefficients:

	Estimate	Std.err	Wald	Pr(> W)	
(Intercept)	-0.2312	0.0798	8.4	0.0038	**
A0	0.9850	0.1022	92.9	<2e-16	***
A1	1.1578	0.0996	135.2	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = independence

Estimated Scale Parameters:

	Estimate	Std.err
(Intercept)	1	0.0514

Number of clusters: 2492 Maximum cluster size: 1

Retrieval Quiz (answer in Complete sentences)

1. Define *sequential exchangeability* and provide a concrete example of when it is plausible.
2. Explain why **positivity** is a property of the data-generating process, not the analyst.
3. What role does the **numerator** of stabilized weights play?
4. Describe two diagnostics you would report to defend your MSM analysis.

Reflection Prompts

- In your own study area, propose a sustained strategy (static or dynamic). Which covariates risk violating positivity, and how would you mitigate that risk?

6 Common Pitfalls & Checks

Common Pitfall

Extreme or unstable weights (heavy right tail) → inflated variance and sensitivity. **Check:** histograms, quantiles; report $\max(sw)$ and $CV(sw)$. **Mitigate:** truncation (e.g., at 1st/99th), coarsen covariates, redefine estimand (narrow regime or support), or switch to g-formula if overlap is non-existent (Hernán & Robins §19–20).

 Common Pitfall

Model misspecification in treatment models → biased MSM even with perfect outcome model neutrality. **Check:** include flexible terms (splines, interactions), consider Super Learner (if allowed), compare alternative specifications (Neal Ch. 10).

 Common Pitfall

Adjusting for mediators in the outcome model (e.g., adding L_1) undermines the marginal target. **Check:** keep MSM outcome models *marginal in $\bar{A}$* and push confounding control into weights.

Check Your Understanding

Why might you **prefer** the g-formula over MSMs when weight tails remain heavy even after truncation?

7 Advanced Teaching Activities & Interactivity

- **Think–Pair–Share:** Given a clinical policy “treat if $L_k > c$,” decide if it’s static or dynamic; identify confounders and mediators at each time; sketch the DAG.
- **Problem-based case:** Simulate a cohort with near-deterministic treatment when L_1 is extreme. Attempt MSM, then redesign the target estimand to satisfy positivity (e.g., restrict to $L_1 \in [0.2, 0.8]$).
- **Spaced/interleaved quiz:** Alternate items asking for (i) weight formulas and (ii) g-formula simulation steps.
- **Self-explanation:** Students write a paragraph comparing the interpretation of β_{A1} in MSM vs the coefficient of A_1 in a fully adjusted outcome regression.
- **Optional Shiny idea:** Live sliders for A_1 assignment strength and L_1 effect; show changes in weight tails and MSM bias.

8 Appendix: Advanced Theory & Derivations (optional)

A.1 Identification Sketch (two times)

Under sequential exchangeability, positivity, and consistency,

$$\mathbb{E}(Y^{a_0, a_1}) = \mathbb{E}\left[\frac{\mathbb{I}(A_0=a_0)\mathbb{I}(A_1=a_1)}{\Pr(A_0=a_0 \mid L_0)\Pr(A_1=a_1 \mid A_0, L_0, L_1)}Y\right].$$

Replacing denominators with estimated propensities yields IPTW. Stabilized weights multiply by numerators $\Pr(A_0=a_0)$ and $\Pr(A_1=a_1 \mid A_0)$ to improve efficiency (Robins, Hernán & Brumback, 2000; Hernán & Robins §19).

A.2 Efficiency and variance

Weighted estimating equations with independence working correlation yield consistent estimates of MSM parameters; robust sandwich variance accounts for misspecification of the working correlation (Liang & Zeger, 1986; see also Hernán & Robins §20).

A.3 Dynamic Regimes

For a dynamic rule $d = (d_0, d_1)$ with $a_k = d_k(\bar{L}_k, \bar{A}_{k-1})$, replace indicators $\mathbb{I}(A_k=a_k)$ with $\mathbb{I}(A_k=d_k(\cdot))$ in the IPTW estimator; the MSM can include summaries of “following the rule” (Cunningham §10; VanderWeele §19).

A.4 Numerical Illustration (mini)

Suppose $\Pr(A_0=1 \mid L_0) = \text{logit}^{-1}(-0.5 + L_0)$; draw 10 individuals; compute denominators and numerators, then SW_i . Summarize $\bar{SW}$, $\max(SW)$, and the truncated proportion. Observe how increasing the $L_1 \rightarrow A_1$ coefficient inflates $\max(SW)$.

References (inline Citations in text)

- Hernán, M. A., & Robins, J. M. (2025). *What If*. (see §§12, 19–20).
- Neal, B. (2020). *Introduction to Causal Inference*. (Chs. 10–11).
- Robins, J. M., Hernán, M. A., & Brumback, B. (2000). Marginal structural models and causal inference in epidemiology.
- VanderWeele, T. (2015). *Explanation in Causal Inference* (Ch. 19).
- Cunningham, S. (2021). *Causal Inference: The Mixtape* (Ch. on panel/time-varying).